Spectral Analysis of Brain Function Network for the Classification of Motor Imagery Tasks

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***Abstract*—In order to deal with the classification for multi-class motor imagery(MI) tasks, a novel approach was presented in this paper. It is different from classical methods which classified the MI task with time-frequency analysis on EEG signals. It employs the brain function network(BFN) as a new characteristic to describe MI tasks. The BFN enlarges the features with respect to traditional time-frequency methods. Unlike analysis of statistical parameters of network such as *average clustering coefficient* (*C*) and the *average pathlength* (*L*), the proposed method employed spectral decomposition performing on BFNs, and considered the eigenvalue vector of threshold matrix as features for classification by SVM. Hence, it is speedy enough to meet the requirement of real-time in BCI-based application systems. The result of experiment demonstrates that proposed method can achieve satisfied accuracy of classification on multi-class MI tasks.**

Keywords-motor imagery;brain function network; classification; BCI;

I. INTRODUCTION

The aim of Brain Computer Interface(BCI) research is to provide a direct link from human intensions, which observed by brain signals, to outside devices such as computers[1]. It becomes a prospective alternative control technology for disabled and able-bodied people. There exist kinds of brain signals to incarnate ones intension, and one of remarkable means is electroencephalogram (EEG) with advantages of cheap equipment, easy measurement and non-invasive.

A commonly used mental strategy for BCIs is motor imagery(MI). The work of Pfurtscheller's team demonstrate that[2-3]: both execution and imagination of the same limb movement activate similar neural structures and result not only in a desynchronization (event-related desynchronization, ERD) of sensorimotor rhythms, but also in a beta rebound (beta event-related synchronization, beta ERS) after termination of the motor imagery task. This team has devoted to MI-BCIs research for more than 15 years. They had built up Graz-I and Graz-II systems to enable subject to imagine his left/right hand, foot or finger to achieve satisfied classification results online. MI-BCIs also have a strong appeal to other researchers over the world. L. Qin and B. He[4] classified the motor imagery tasks by the time-frequency analysis on EEG signals based on wavelet transformation. Additionally, they also traced EEG signal sources for the motor imagery tasks, and took the source positions as features for classification[5]. Yamawaki[6] in Japan built a time-frequency model to study the EEG signal, and achieved satisfied results for classification. X. R. Gao in Qinghua University utilized the energy of specific frequency band as feature to study the classification of motor imagery[7]. L.Q. Zhang proposed a multi-linear formulation of the CSP, termed as Tensor CSP or Common Tensor Discriminant Analysis (CTDA) for high-order tensor data to improve the classification accuracy of multi-class motor imagery EEG[8].

However, all the above mentioned methods focused on the specific region of brain that is called motor cortex. Therefore, their researches are only based on isolated brain region, but ignoring the overall coordination of brain function. In fact, in order to accomplish a very simple task, the brain needs multi function regions to interact and coordinate with each other as a network[9-10]. On the other hand, above methods usually only can deal with 2-class tasks due to insufficient features. Consequently, we proposed a novel classification method for multi-class motor imagery tasks with brain function network. It takes the brain function network as an original point in the data space, then transformed it into a new feature space, finally, support vector machine(SVM) is employed to classify the intention of brain.

II. BRAIN FUNCTION NETWORK

A. *Overview of Brain Function Network*

The brain is inherently a dynamic system, in which the traffic between regions, during any cognitive tasks even at rest, creates and reshapes continuously complex functional networks of correlated dynamics[9-10]. Brain function network (BFN) is an effective mean for describing the coordination among different brain areas. It is also can be used as another kind of measurement for relation and causality for disparate brain regions. Hence more and more scientists pay close attention to BFN. Bullmore has discovered the property of ‘‘small-word’’ for BFN, and demonstrated that it is a high active model to portray the Brain areas[10]. F. Babiloni has employed BFN to analyze its two important attributes(cluster coefficient and average path-length of the network) for evaluating the effect on subjects[11] by TV commercial materials.

B. Constructing Brain Function Network

According to the method to extract normal functional network, BFN can be constructed from EEG signals, and analyzed for motor imagery tasks.

In our paper, the EEG signals are collected by fifteen scalp electrodes (so-called “channels’’). At each time window, the activity of channel x is denoted as vector X, and $\overline{X}$ represents its temporal average. We define that two channels are functionally connected if their temporal correlation exceeds a positive threshold value rc, regardless of their anatomical connectivity [9-10]. There are several methods to measure the function connectivity, i.e., correlation, coherence, and phase synchronization. Specially, we employ the Pearson correlation coefficient between any pair of channels, x and y, as the degree of function connectivity. It can be denoted as following:

$$r(x,y)=\frac{\sum_{i=1}^{n}(X_i-\overline{X})(Y_i-\overline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i-\overline{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i-\overline{Y})^2}} \tag{1}$$

Where n is the number of samples at a time window.

The threshold matrix is transformed from the correlation matrix. At each time window, each electrode is regarded as a vertex of the network. If the correlation value $r(x,y)$ is above the threshold rc, then $r(x,y)$ equals to 1, else $r(x,y)$ equals to 0. Consider the correlation value $r(x,y)$ is 1, we put an edge between corresponding two vertices(channels), otherwise they are disconnected. Therefore, according to the threshold matrix, we can connect to the corresponding vertices via edges and then extract the network. Figure 1 shows how to construct BFN during a certain motor imaginary tasks.

C. Feature extraction for BFN

1) Graph-based Feature extraction

As soon as the network for certain cognitive task has been extracted, graph theory can be immediately used to analyze the brain functional network, such as the *average clustering coefficient* (C) and the *average pathlength* (L)[9]. These two parameters are important extracted features from network. Clustering (C) is the fraction of connections between the topological neighbors of a vertex with respect to the maximum possible. If vertex i has degree k_i which is the number of vertices linked to vertex i, then the maximum number of links between the k_i neighbors is k_i (k_i -1)/2.

The clustering coefficient of a vertex i with degree k_i is defined as

$$C(i)=2E_i/(k_i\times(k_i-1)) \tag{2}$$

Where E_i is the number of edges amongst the neighbors of vertex i, So the average clustering coefficient is

$$C=1/N\sum_i C(i) \tag{3}$$

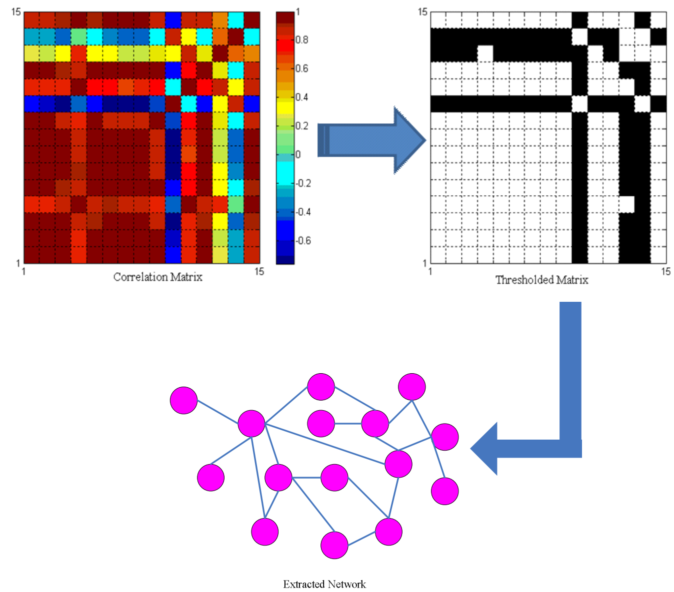

Figure 1. Extracting functional networks from the correlation matrix

The path length (L) between two vertices is the minimum number of links necessary to connect both vertices. These statistical parameters are widely used in the analysis of certain cognitive tasks. The disadvantage of this approach is complex and time consumed, so it is usually offline analyzed. However, real-time is important for the applications of MI-BCIs, so we need search for a new and fast approach of feature extraction.

2) Spectral analysis of BFN

From fig.1, we can conduct that Brain function network is extracted from threshold matrix. Thus, analyzing the threshold matrix is equivalent to analyzing the network. Let the threshold symmetry matrix $\boldsymbol{R}$ be

$$\boldsymbol{R}=\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1N} \\ r_{21} & r_{22} & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N1} & r_{N2} & \cdots & r_{NN} \end{bmatrix} \tag{4}$$

Where the entity r_{ij} is either 1 or 0, and $r_{ij}=r_{ji}$. For $\boldsymbol{R}$, it can be eigenvalue decomposed(EVD) as $\boldsymbol{R}=\boldsymbol{P\Lambda P}^{-1}$,where $\boldsymbol{P}$ and $\boldsymbol{\Lambda}$ are matrixes composed by eigenvectors and eigenvalues of $\boldsymbol{R}$ respectively. After the linear elementary transformation on $\boldsymbol{R}$, we can get a block diagonal matrix:

$$\boldsymbol{R}=\begin{bmatrix} \boldsymbol{R}_1 & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{R}_2 & \cdots & \boldsymbol{0} \\ \vdots & \cdots & \ddots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{R}_s \end{bmatrix} \tag{5}$$

As $\boldsymbol{R}=diag\{\boldsymbol{R}_1,\boldsymbol{R}_2\cdots\boldsymbol{R}_s\}$ where $s\leq N$, $\boldsymbol{\Lambda}=diag\{\boldsymbol{\Lambda}_1,\boldsymbol{\Lambda}_2\cdots\boldsymbol{\Lambda}_s\}$, and $\boldsymbol{\Lambda}_i=\lambda_i\boldsymbol{E}$,where λ_i is the eigenvalue of sub-matrix $\boldsymbol{R}_i$. So

$$\begin{aligned} \boldsymbol{R}\boldsymbol{A}\boldsymbol{R}^{-1} &= diag\{\boldsymbol{R}_1(\lambda_1\boldsymbol{E})\boldsymbol{R}_1^{-1}, \boldsymbol{R}_2(\lambda_2\boldsymbol{E})\boldsymbol{R}_2^{-1}, \cdots \boldsymbol{R}_s(\lambda_s\boldsymbol{E})\boldsymbol{R}_s^{-1}\} \\ &= diag\{\lambda_1\boldsymbol{R}_1\boldsymbol{E}\boldsymbol{R}_1^{-1}, \lambda_2\boldsymbol{R}_2\boldsymbol{E}\boldsymbol{R}_2^{-1}, \cdots \lambda_s\boldsymbol{R}_s\boldsymbol{E}\boldsymbol{R}_s^{-1}\} \\ &= diag\{\lambda_1\boldsymbol{E}, \lambda_2\boldsymbol{E}, \cdots \lambda_s\boldsymbol{E}\} \\ &= diag\{\boldsymbol{\Lambda}_1, \boldsymbol{\Lambda}_2, \cdots \boldsymbol{\Lambda}_s\} = \boldsymbol{\Lambda} \end{aligned} \tag{6}$$

From Eq.(6), we know that $\boldsymbol{\Lambda}$ can also be eigenvalue decomposed by its original corresponding symmetry matrix $\boldsymbol{R}$. In other words, $\boldsymbol{\Lambda}$ is another kind of pattern to denote symmetry matrix $\boldsymbol{R}$. Finally, what is $\boldsymbol{\Lambda}$? It is the set of eigenvalues of $\boldsymbol{R}$. Therefore, we can translate 2-dimension matrix $\boldsymbol{R}$ into 1-dimension vector $\vec{\boldsymbol{\lambda}} = [\lambda_1, \lambda_2 \cdots \lambda_s]^{\mathrm{T}}$.

Hence, the eigenvalue vector of threshold matrix, can be taken as extracted feature, which is used for the classification of motor imagery tasks.

III. SVM FOR CLASSIFICATION

A. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a new method for the classification of both linear and nonlinear data[12]. In a nutshell, it is an algorithm that works as follows. Firstly, it uses a nonlinear mapping to transform the original training data into a higher dimension. Then, it searches for the linear optimal separating hyperplane(that is, a “decision boundary” separating tuples of one class from another) within this new dimension. As an appropriate nonlinear mapping to a sufficiently high dimension, data with two classes can always be separated by a hyperplane.

In this paper, each eigenvalue vector is regarded as a sample in a new space, the category of motor imagery task can be regarded as assigning the sample label. LIBSVM toolbox which supports multi-class classification is employed to classify the intention of brain. Specially, polynomial kernel of degree q (Polynomial) is taken as the type of kernel function, It can be denoted as

$$K(\boldsymbol{x}, \boldsymbol{x}_i) = (\boldsymbol{x} \cdot \boldsymbol{x}_i + 1)^q \tag{7}$$

To construct an optimal hyperplane, SVM employees an iterative training algorithm, which is used to minimize an error function as follow:

$$\min \quad \frac{1}{2}\| w \|^2 + C\sum_{i=1}^{l} | y_i - g(\boldsymbol{x}_i) |_\varepsilon \tag{8}$$

where $| y - g(\boldsymbol{x}) |_\varepsilon = \max(0, | y - g(\boldsymbol{x}) | - \varepsilon)$, w is the normal vector of the linear function; $g(\boldsymbol{x}) = w \cdot \boldsymbol{x} + b$ is the linear discriminant function; $\boldsymbol{x}$ is the eigenvalue vector, y is its corresponding label; C is a tradeoff coefficient between two terms; i is the index of training samples. In function (8), the front part is a constraint on the linear functions and the second half part is a constraint on the training error.

Due to high dimension, it will be a difficult problem to solve function (8) directly. According to the optimization theory, it can be turned into its dual problem as follows:

$$\begin{aligned} &\max \quad W(\alpha) = \sum_{i=1}^{l} y_i \alpha_i - \varepsilon \sum_{i=1}^{l} |\alpha_i| - \frac{1}{2}\sum_{i,j=1}^{l} \alpha_i \alpha_j K(\boldsymbol{x}_i, \boldsymbol{x}_j) \\ &s.t \quad \sum_{i=1}^{l} \alpha_i = 0 \quad -C \le \alpha_i \le C \quad i = 1, 2 ... l \end{aligned}$$

(9)

Where α is called lagrange multiplier. Once we get the lagrange multiplier and the intercept b^*, the estimation can be denoted as:

$$g(\boldsymbol{x}) = \sum_{i=1}^{l} \alpha_i \cdot K(\boldsymbol{x}_i, \boldsymbol{x}) + b^* \tag{10}$$

Where $\boldsymbol{x}_i$ is the training sample, and $\boldsymbol{x}$ is the testing sample.

IV. EXPERIMENTS

We collected EEG datasets in our laboratory, the experiment used a amplifier provide by Tsinghua University to collect EEG signals with fifteen scalp electrodes, namely, the dimension of the feature space is 15. The sampling frequency of amplifier is 1000Hz. And subjects are male and right-handed with ages range in 25-30. Subjects are also sound in body and mind. They are also selected for experiments, because motor imagery task differs from subject to subject. Subjects should be well trained to produce good enough EEG signals. In our lab, only a few subjects could master the motor imagery task. The datasets are collected in different days. Three of them are 2-class motor imagery tasks, i.e., imaging left hand and right hand, the rest are 4-class tasks, i.e., imaging left hand, right hand, both hands and both feet. The time window in constructing brain function network is 256ms, and the threshold $rc = 0.5$.

In the experiment, we first compare the performance for two different methods of feature extraction, one is energy spectrum based on FFT, and the other is the eigenvalue vector based on BFN. Due to the rare features for the index of energy spectrum based on FFT, it only can classify 2-state tasks. Hence 3 datasets with 2-state tasks (imaging left and right) are tested both on FFT-based and BFN-based feature extraction methods, but with the same classification method i.e., nearest neighbor(NN). The results are shown in table 1. From the Table 1, we can conclude that a higher accuracy can be obtained by BFN than FFT.

Table 1. 2-class tasks experiment results with FFT and BFN

2-class	FFT	BFN
Dataset1	0.7258	0.9429
Dataset2	0.8083	0.8157
Dataset3	0.8356	0.8436

Besides, we also carry out another experiments on 4 datasets, with the same feature extraction method i.e., BFN, but with two different classifications, one is nearest neighbor (NN) and the other is support vector machine(SVM). The results of Table 2 demonstrate that BFN+SVM is generally better than BFN+NN in terms of classification accuracy, about 10% gap. Furthermore, it is also very important that it provides a new approach of classification for multi-class motor imagery tasks.

Table 2. 4-class tasks experiment results with BFN plus NN and SVM respectively

4-class	training size	test size	Classification accuracy	
			NN	SVM
Dataset4	1108	1108	0.8366	0.9639
Dataset5	900	900	0.5367	0.6567
Dataset6	900	900	0.5988	0.6447
Dataset7	945	945	0.5170	0.6444

In order to illustrate the method of BFN+SVM more clearly, we employ the proposed approach on a certain 4-class EEG dataset. The size of dataset is with 891 samples(networks). The experiment result is as fig.2 shows. The implications of *class index* in Fig.2 are as following: 1 means imaging left hand, 2means right hand, 3 means both hands, and 4 means both feet. It is obvious that there are almost no errors occurred when classifying hand-motor imagery, but misclassified a little both feet samples into classes of both hands and left hand.

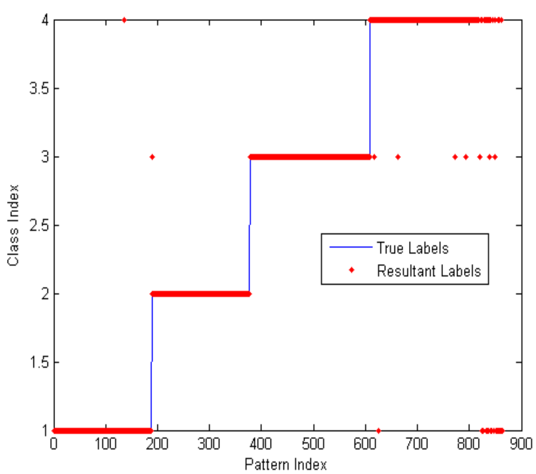

Fig.2 Classification result on 4-class dataset by BFN+SVM

It should be pointed out that the value of *rc* and some parameters of SVM will also influent classification results, as subject depended as well. And, all datasets are continuing task which each imaging state lasts for about 2 minutes.

V. CONCLUSION

In this paper, a novel approach of classification on motor imagery(MI) tasks was presented. It is different from classical methods to classify the MI task with time-frequency analysis on EEG signals. It employs the brain function network(BFN) as a new characteristic to depict MI tasks. We extract a vector to denote the BFN in the proposed method. Because the feature is a multi-dimension vector, so it is information sufficient to distinguish 4-state MI tasks. Experiment results also reveal that SVM is much better than NN for classification on BFN feature vector. Furthermore, it is simple enough to meet the requirement of real-time in BCI-based application system.

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